Week-2
Mathematics for Data Science - 2
Solutions of system of linear equations
Practice Assignment

1. Consider the following systems of equations and choose the correct option.

System I:
$$\begin{aligned} -x + 2y - 2z &= 2 \\ 2x + z &= -1 \\ x - 3y + z &= 3 \end{aligned}$$

System II:
$$\begin{aligned} -2x + y + z &= 0 \\ \frac{3}{2}x + 2y - z &= -2 \\ 3x + 4y - 2z &= 5 \end{aligned}$$

System III:
$$\begin{aligned} x + 3z &= -5 \\ -\frac{2}{5}x - \frac{1}{5}y - 2z &= 3 \\ 2x + y + 10z &= -15 \end{aligned}$$

- ◯ **Option 1:** System I has a unique solution.
- ◯ Option 2: System II has infinitely many solutions.
- ◯ **Option 3:** System II has no solution.
- ◯ Option 4: System III has no solution.

Solution:
System I: The given system of linear equations is

$$\begin{aligned} -x + 2y - 2z &= 2 \\ 2x + z &= -1 \\ x - 3y + z &= 3 \end{aligned}$$

The matrix representation of this system is $AX = B$
where $A = \begin{bmatrix} -1 & 2 & -2 \\ 2 & 0 & 1 \\ 1 & -3 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ *and* $B = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$

The augmented matrix for this system is $[A|B] = \left[\begin{array}{ccc|c} -1 & 2 & -2 & 2 \\ 2 & 0 & 1 & -1 \\ 1 & -3 & 1 & 3 \end{array}\right]$

$$\xrightarrow{R_2+2R_1, R_3+R_1} \left[\begin{array}{ccc|c} -1 & 2 & -2 & 2 \\ 0 & 4 & -3 & 3 \\ 0 & -1 & -1 & 5 \end{array}\right]$$

$$\xrightarrow{R_3+1/4R_2} \left[\begin{array}{ccc|c} -1 & 2 & -2 & 2 \\ 0 & 4 & -3 & 3 \\ 0 & 0 & -\frac{7}{4} & \frac{23}{4} \end{array}\right]$$

By back substitution, we have $-\frac{7}{4}z = \frac{23}{4} \implies z = -\frac{23}{7}$

$4y - 3z = 3 \implies 4y = 3 - 3(\frac{23}{7}) \implies y = -\frac{48}{28} = -\frac{12}{7}$

$-x + 2y - 2z = 2 \implies -x - \frac{24}{7} + \frac{46}{7} = 2 \implies x = \frac{8}{7}$

Hence the system I has a unique solution.
So, option 1 is correct.

System II: The given system of linear equations is

$$\begin{aligned} -2x + y + z &= 0 \\ \frac{3}{2}x + 2y - z &= -2 \\ 3x + 4y - 2z &= 5 \end{aligned}$$

The matrix representation of this system is $AX = B$

where $A = \begin{bmatrix} -2 & 1 & 1 \\ \frac{3}{2} & 2 & -1 \\ 3 & 4 & -2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ *and* $B = \begin{bmatrix} 0 \\ -2 \\ 5 \end{bmatrix}$

The augmented matrix for this system is $[A|B] = \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ \frac{3}{2} & 2 & -1 & -2 \\ 3 & 4 & -2 & 5 \end{array}\right]$

$$\xrightarrow{R_2+\frac{3}{4}R_1} \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 0 & \frac{11}{4} & \frac{-1}{4} & -2 \\ 3 & 4 & -2 & 5 \end{array}\right]$$

$$\xrightarrow{R_3+\frac{3}{2}R_1} \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 0 & \frac{11}{4} & \frac{-1}{4} & -2 \\ 0 & \frac{11}{2} & \frac{-1}{2} & 5 \end{array}\right]$$

$$\xrightarrow{R_3-2R2} \left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 0 & \frac{11}{4} & \frac{-1}{4} & -2 \\ 0 & 0 & 0 & 9 \end{array}\right]$$

The last equation in the reduced system says $0 = 9$, which is absurd.

Hence the system II has no solution.
Option 3 is correct.

System III: The given system of linear equations is

$$\begin{aligned} x + 3z &= -5 \\ -\frac{2}{5}x - \frac{1}{5}y - 2z &= 3 \\ 2x + y + 10z &= -15 \end{aligned}$$

The matrix representation of this system is $AX = B$

where $A = \begin{bmatrix} 1 & 0 & 3 \\ -\frac{2}{5} & -\frac{1}{5} & -2 \\ 2 & 1 & 10 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ *and* $B = \begin{bmatrix} -5 \\ 3 \\ -15 \end{bmatrix}$

The augmented matrix for this system is $[A|B] = \left[\begin{array}{ccc|c} 1 & 0 & 3 & -5 \\ -\frac{2}{5} & -\frac{1}{5} & -2 & 3 \\ 2 & 1 & 10 & -15 \end{array}\right]$

$$\xrightarrow{R_2 - \frac{-2}{5}R_1} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -5 \\ 0 & -\frac{1}{5} & -\frac{4}{5} & 1 \\ 2 & 1 & 10 & -15 \end{array}\right]$$

$$\xrightarrow{R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -5 \\ 0 & -\frac{1}{5} & -\frac{4}{5} & 1 \\ 0 & 1 & 4 & -5 \end{array}\right]$$

$$\xrightarrow{R_3 - (-5)R2} \sim \left[\begin{array}{ccc|c} 1 & 0 & 3 & -5 \\ 0 & -\frac{1}{5} & -\frac{4}{5} & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Since one of the variables is independent, the system has infinite number of solutions.

Hence the system III has infinitely many solutions.
No option is correct for System III.

2. There are two laptop manufacturers, one is at Adyar and the other is at Tambaram. Suppose the production costs (in crore of ₹) at Adyar and Tambaram are represented by the equations $A(x) = a_1x^2 + b_1x + c_1$ and $T(x) = a_2x + c_2$, respectively, where x represents the number (in hundreds) of laptops produced. At Adyar, the initial investment is known to be ₹3 crore, and the production costs for manufacturing 100 (i.e., $x = 1$) and 300 laptops (i.e., $x = 3$) are ₹4 crore and ₹12 crore, respectively. At Tambaram, the production costs for manufacturing 100 and 200 laptops are ₹6 crore and ₹7 crore, respectively. Suppose, Parveena and Amenla need new laptops for their start up companies. Parveena needs 500 laptops and Amenla needs 150 laptops. Both of them want their laptops with minimum production cost. Choose the correct option from the given set of options below.

- ◯ **Option 1:** Cost of production of 200 laptops at each Adyar and Tambaram is ₹7 crore.
- ◯ Option 2: Cost of production of 100 laptops at each Adyar and Tambaram is ₹5 crore.
- ◯ Option 3: Parveena should place her order at Adyar and Amenla should place her order at Tambaram to avail the minimum production cost.
- ◯ **Option 4:** Parveena should place her order at Tambaram and Amenla should place her order at Adyar to avail the minimum production cost.

Solution:

Suppose the production costs (in crore of ₹) at Adyar and Tambaram are represented by the equations

$$A(x) = a_1x^2 + b_1x + c_1$$

and

$$T(x) = a_2x + c_2,$$

respectively, where x represents the number (in hundreds) of laptops produced.

Given that the initial investment at Adyar is ₹3 crore, i.e., if we substitute $x = 0$ in $A(X)$ we get $c_1 = 3$. Thus

$$A(x) = a_1x^2 + b_1x + 3.$$

Also given that at Adyar, the production cost for manufacturing 100 laptops (i.e., x=1) is ₹4 crore. i.e.,

$$4 = a_1 + b_1 + 3 \implies a_1 + b_1 = 1.$$

Production cost for manufacturing 300 laptops (i.e., $x = 3$) is ₹12 crore.

$$9a_1 + 3b_1 + 3 = 12 \implies 3a_1 + b_1 = 3$$

Solving the above two equations we get, $a_1 = 1, b_1 = 0$.

Hence $A(x) = x^2 + 3$.

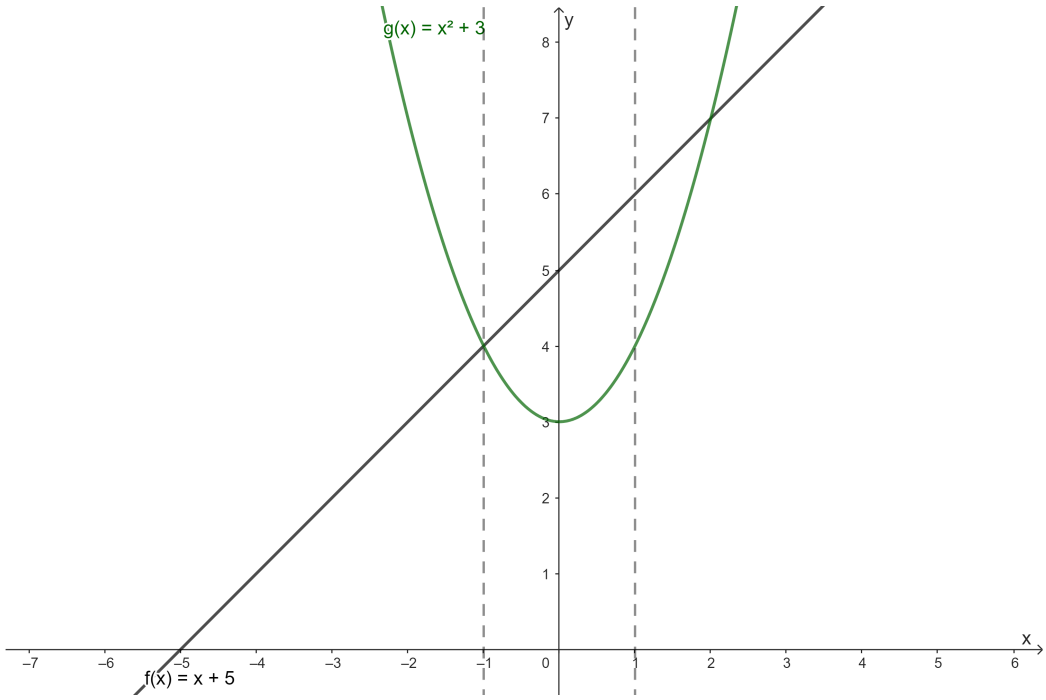
Similarly, at Tambaram, the production cost for manufacturing 100 laptops (i.e., x=1) is ₹6 crore, so $6 = a_2 + c_2$ and for 200 laptops (i.e., x=2) the production cost is ₹7 crore, so $7 = 4a_2 + c_2$. Solving these we get, $a_2 = 1$ and $c_2 = 5$.

Hence $T(x) = x + 5$.

The following is the graph of the function $A(x) = x^2 + 3$ and $T(x) = x + 5$.

Production cost for manufacturing 500 laptops at Adyar is $A(5) = 5^2 + 3 = 28$ crore rupees and at Tambaram is $T(5) = 5 + 5 = 10$ crore rupees.

Production cost for manufacturing 150 laptops at Adyar is $A(1.5) = 1.5^2 + 3 = 5.25$ crore rupees and at Tambaram is $T(1.5) = 5 + 5 = 6.5$ crore rupees.



So the production cost for manufacturing 500 laptops is minimum at Tambaram and the production cost for manufacturing 150 laptop is minimum at Adyar.

So Parveena should place her order at Tambaram and Amenla should place her order at Adyar to avail minimum production cost.

Hence option 4 is true.

Now the production cost for 200 laptop at Adyar is $A(2) = 2^2 + 3 = 7$ crore and at Tambaram is $T(2) = 2 + 5 = 7$ crore.

Hence option 1 is true.

Similarly the production cost for 100 laptop at Adyar is $A(1) = 1^2 + 3 = 4$ crore and at Tambaram is $T(1) = 1 + 5 = 6$ crore.

Hence option 2 is not true.

3. Choose the set of correct options.

   ◯ **Option 1:** If A is an upper triangular 3×3 matrix, then the adjoint matrix of A is also an upper triangular matrix.

   ◯ **Option 2:** If A is an invertible upper triangular 3×3 matrix, then the inverse matrix of A is also an upper triangular matrix.

- ◯ Option 3: Let A be an arbitrary real 3×3 matrix. If C is the adjoint matrix of A, then C is also the adjoint matrix of A^T.
- ◯ Option 4: C_{jk} denotes the cofactor with respect to the j-th row and the k-th column of a 3×3 matrix A. If another matrix B is obtained from A by replacing the j-th row of A with $\begin{bmatrix} 3 & 0 & 0 \end{bmatrix}$, then $det(B) = 3C_{jk}$
- ◯ Option 5: If A is an invertible 3×3 matrix and $C = adj(adj(A))$, then $det(C) = det(A)^9$

Solution:

Option 1: Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$ be an upper triangular 3×3 matrix, then the adjoint matrix of A is $adj(A) = \begin{bmatrix} a_{22}.a_{33} & -a_{12}.a_{33} & a_{12}.a_{23} - a_{22}.a_{13} \\ 0 & a_{11}.a_{33} & -a_{11}.a_{23} \\ 0 & 0 & a_{11}.a_{22} \end{bmatrix}$ which is also an upper triangular matrix.

Hence option 1 is true.

Option 2: We know that $A^{-1} = \frac{adj(A)}{det(A)}$

So $A^{-1} = \frac{1}{det(A)} . \begin{bmatrix} a_{22}.a_{33} & -a_{12}.a_{33} & a_{12}.a_{23} - a_{22}.a_{13} \\ 0 & a_{11}.a_{33} & -a_{11}.a_{23} \\ 0 & 0 & a_{11}.a_{22} \end{bmatrix}$

which is also an upper triangular matrix.

Hence option 2 is true.

Option 3:

Let $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ then $\text{adj}(A) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Now, $A^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ then $\text{adj}(A^T) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$.

Hence $adj(A) \neq adj(A^T)$.

Hence option 3 is not true.

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Let $j = 2$ and $k = 2$ then $C_{22} = 1$. As B is obtained from A by replacing the jth row of A with $[3, 0, 0]$, for the case $j = 2$ we have, $B = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

So $det(B) = 0 \neq 3C_{22}$.

Hence option 4 is not true.

Option 5: **First method:** Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then $adj(A) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $adj(adj(A)) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 18 \end{bmatrix}$. We know that the determinant of a diagonal matrix is just product of diagonal elements so $det(A) = 6$ and $det(adj(adj(A))) = 6 \times 12 \times 18 = 6^4$.

Hence $det(adj(adj(A))) = 6^4 \neq det(A)^9 = 6^9$.

Second Method:

Given $C = adj(adj(A))$. Let $m = det(A)$, we have $A^{-1} = \frac{adj(A)}{det(A)}$ and so $adj(A) = A^{-1}det(A)$.

$$\begin{aligned}
det(C) =& det(adj(adj(A))) \\
=& det(adj(det(A)A^{-1})) \\
=& det(adj(mA^{-1})) \\
=& det(det(mA^{-1})(mA^{-1})^{-1}) \\
=& det(m^3 det(A^{-1})\frac{1}{m}(A^{-1})^{-1}) \\
=& det(m^3 det(A)^{-1}\frac{1}{m}(A)) \\
=& det(m^3 (m)^{-1}\frac{1}{m}(A)) \\
=& det(m^3 \frac{1}{m}\frac{1}{m}(A)) \\
=& det(mA) \\
=& m^3 det(A) \\
=& m^3 m = m^4 \\
det(C) =& det(A)^4
\end{aligned}$$

Hence option 5 is not true.

4. Choose the correct set of options based on the matrices given in Table M2W2PT1.

Column A	Column B
$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	$B_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$B_2 = \begin{bmatrix} -\frac{2}{3} & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \\ -\frac{1}{3} & 0 & 0 \end{bmatrix}$
$A_3 = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix}$	$B_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
$A_4 = \begin{bmatrix} 0 & 0 & -3 \\ 1 & 0 & -2 \\ 0 & 1 & -1 \end{bmatrix}$	$B_4 = \begin{bmatrix} -2 & 1 & 0 \\ -3 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$

Table: M2W2PT1

- ◯ **Option 1:** A_1 and B_3 are inverses to each other.
- ◯ Option 2 : A_1 and B_1 are inverses to each other.
- ◯ **Option 3 :**A_2 and B_1 are inverses to each other.
- ◯ **Option 4:** A_3 and B_4 are inverses to each other.
- ◯ Option 5: A_2 and B_3 are inverses to each other.
- ◯ Option 6: A_3 and B_2 are inverses to each other.
- ◯ Option 7: A_4 and B_4 are inverses to each other.
- ◯ **Option 8:** A_4 and B_2 are inverses to each other.
- ◯ Option 9: A_2 and A_3 have different reduced row echelon form.
- ◯ Option 10: A_1 and A_2 have different reduced row echelon form.
- ◯ **Option 11:** All the matrices in column A have the same reduced row echelon form and that is the identity matrix of order 3.
- ◯ Option 12: All the matrices in column A have the same reduced row echelon form but that is not the identity matrix of order 3.

Solution: Given $A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$|A_1| = 1[-1] = -1$

$adj(A_1) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

So ${A_1}^{-1} = \frac{adj(A_1)}{|A_1|} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = B_3.$

Hence A_1 and B_3 are inverses to each other. Hence option 1 is correct.

Given $A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$|A_2| = 1[-1] = -1$

$adj(A_2) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

So ${A_2}^{-1} = \frac{adj(A_2)}{|A_2|} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = B_1.$

Hence A_2 and B_1 are inverses to each other. Hence option 3 is correct.

Given $A_3 = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix}$

$|A_3| = -1[1] = -1$

$adj(A_3) = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$

So ${A_3}^{-1} = \frac{adj(A_3)}{|A_3|} = \begin{bmatrix} -2 & 1 & 0 \\ -3 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} = B_4.$

Hence A_3 and B_4 are inverses to each other. Hence option 4 is correct.

Given $A_4 = \begin{bmatrix} 0 & 0 & -3 \\ 1 & 0 & -2 \\ 0 & 1 & -1 \end{bmatrix}$

$|A_4| = -3[1] = -3$

$$adj(A_4) = \begin{bmatrix} 2 & -3 & 0 \\ 1 & 0 & -3 \\ 1 & 0 & 0 \end{bmatrix}$$

So ${A_4}^{-1} = \frac{adj(A_3)}{|A_3|} = \begin{bmatrix} -\frac{2}{3} & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \\ -\frac{1}{3} & 0 & 0 \end{bmatrix} = B_2.$

Hence A_4 and B_2 are inverses to each other. Hence option 8 is correct.

Now $A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

which is the row reduced echelon form of A_1.

$A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{(-1)R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is the row reduced echelon form of A_2.

$A_3 = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & -1 \\ 0 & 1 & -3 \end{bmatrix}$

$\xrightarrow{(-1)R_2} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_1+2R_2, R_3+3R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$\xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is the row reduced echelon form of A_3.

$A_4 = \begin{bmatrix} 0 & 0 & -3 \\ 1 & 0 & -2 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & -3 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{\frac{-1}{3}R_2} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_1+2R_2, R_3+R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is the row reduced echelon form of A_4.

Clearly all matrices in Column A have same reduced row echelon form which is identity matrix of order 3.

Hence option 11 is correct.

5. The sum of the diagonal entries of the row reduced echelon form of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -1 & 2 & 0 \end{bmatrix}$ is [Answer: 2]

Solution:
$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -1 & 2 & 0 \end{bmatrix}$

Applying row transformation $R_3 \to R_3 - R_2$, $A \sim \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -2 & 0 & 0 \end{bmatrix}$,

$R_3 \to R_3 + 2R_1,\ R_2 \to R_2 - R_1\ A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$,

$R_2 \to \frac{R_2}{2}\ A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

which is the required row reduced echelon form of the matrix A, hence sum of the diagonal elements is 2.

6. A gym trainer suggested Pranjal to include banana, mozzarella cheese, and avocado in his daily diet, for his fitness. In 1 banana, there are 10 units of protein, 50 units of carbohydrate, and 0 unit of fat. In $\frac{1}{2}$ cup mozzarella cheese, there are 1 units of protein, 20 units of carbohydrate and 1 unit of fat. In 1 avocado there are 3 units of protein, 10 units of carbohydrate, and 10 units of fat. Suppose the calories intake from 1 banana, $\frac{1}{2}$ cup mozzarella cheese, and 1 avocado are 105, 90 and 115, respectively. If the gym trainer suggested Pranjal to take 18 units of protein, 110 units of carbohydrate, and 22 units of fat by taking only these three items, then find out the calories intake by Pranjal each day from these three items only. [Answer: 530]

Solution: From the given information we have,

	Protein	Carbohydrate	Fat	Calories
1 Banana	10	50	0	105
$\frac{1}{2}$ cup mozzarella cheese	1	20	1	90
1 avocado	3	10	10	115

Table: M2W2PT2

Let, to take 18 units of protein, 110 units of carbohydrate, and 22 units of fats, Pranjal takes x numbers of banana, y numbers of $\frac{1}{2}$ cup mozzarella cheese and z number of avocado, then 18 units of protein can get by taking $10x + y + 3z = 18$. Similarly for carbohydrate and fat we have $50x + 20y + 10z = 110 \implies 5x + 2y + z = 11$ and $0x + 1y + 10z = 22$.

These represent the system of three equations in three unknowns, can be represented by the matrix as $AX = b$, where $A = \begin{bmatrix} 10 & 1 & 3 \\ 5 & 2 & 1 \\ 0 & 1 & 10 \end{bmatrix}$,

$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ *and* $b = \begin{bmatrix} 18 \\ 11 \\ 22 \end{bmatrix}$.

The augmented matrix for this system is $[A|b] = \left[\begin{array}{ccc|c} 10 & 1 & 3 & 18 \\ 5 & 2 & 1 & 11 \\ 0 & 1 & 10 & 22 \end{array}\right]$

and the reduced echelon form is $\left[\begin{array}{ccc|c} 1 & \frac{1}{10} & \frac{3}{10} & \frac{9}{5} \\ 0 & 1 & 10 & 22 \\ 0 & 0 & 1 & 2 \end{array}\right]$ From the above matrix we get, $x = 1$, $y = 2$ and $z = 2$. So Pranjal will take 2 bananas, only one $\frac{1}{2}$ cup of mozzarella cheese and 2 avocado.

Calories intake from 1 banana, one $\frac{1}{2}$ cup of mozzarella cheese and 1 avocado are 105, 90 and 115 respectively.

Hence total calories taken by Pranjal are $1\times105+2\times90+2\times115 = 105+180+230 = 515$.

7. Consider the system of linear equations $Ax = b$, where $A = \begin{bmatrix} 2 & a & 3 \\ a & -2 & -1 \\ -1 & a & 0 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ \frac{5}{4} \\ x_3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ a \\ 1 \end{bmatrix}$. The solution x is partially known. What is the value of a if $a > 1$ is given? [Answer: 2]

Solution: Given, $A = \begin{bmatrix} 2 & a & 3 \\ a & -2 & -1 \\ -1 & a & 0 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ \frac{5}{4} \\ x_3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ a \\ 1 \end{bmatrix}$.

Observe that we $x_2 = \frac{5}{4}$.

$|A| = \begin{vmatrix} 2 & a & 3 \\ a & -2 & -1 \\ -1 & a & 0 \end{vmatrix}$.

Expanding along R_3, we get

$|A| = -1(-a+6) - a(-2-3a) + 0 = 3a^2 + 3a - 6$

Replacing the second column of A with b we get,

$A_{x_2} = \begin{vmatrix} 2 & 1 & 3 \\ a & a & -1 \\ -1 & 1 & 0 \end{vmatrix}$.

Expanding along R_3, we get

$|A_{x_2}| = -1(-1-3a) - 1(-2-3a) = 1 + 3a + 2 + 3a = 6a + 3$

Now using Cramer's rule, $x_2 = \frac{5}{4} = \frac{|A_{x_2}|}{|A|} = \frac{6a+3}{3a^2+3a-6}$.
This is well defined as $|A| \neq 0$ as $a > 1$.

$$\begin{aligned}
& \frac{5}{4} = \frac{6a+3}{3a^2+3a-6} \\
\implies & 5(3a^2+3a-6) = 4(6a+3) \\
\implies & 15a^2 + 15a - 30 = 24a + 12 \\
\implies & 5a^2 + 5a - 10 = 8a + 4 \\
\implies & 5a^2 - 3a - 14 = 0 \\
\implies & 5a^2 - 10a + 7a - 14 = 0 \\
\implies & 5a(a-2) + 7(a-2) = 0 \\
\implies & (a-2)(5a+7) = 0 \\
\implies & a = 2, a = -\frac{7}{5}
\end{aligned}$$

As $a > 1$, the possible value of a is 2.

Comprehension Type Question:

In genetics, a classic example of dominance is the inheritance of seed shape (pea shape) in peas. Peas may be round (associated with genotype R) or wrinkled (associated with genotype r). In this case, three combinations of genotypes are possible: RR, rr, and Rr. The RR individuals have round peas and the rr individuals have wrinkled peas. In Rr individuals the R genotype masks the presence of the r genotype, so these individuals also have round peas. Thus, the genotype R is completely dominant to genotype r, and genotype r is recessive to genotype R. First, assume the crossing of RR with RR. This always gives the genotype RR, therefore the probabilities of an offspring to be RR, Rr, and rr respectively are equal to 1, 0, and 0. Second, assume crossing of Rr with RR. The offspring will have equal chances to be of genotype RR and genotype Rr, therefore the probabilities of RR, Rr, and rr repectively are 1/2, 1/2, and 0. Third, consider crossing of rr with RR. This always results in genotype Rr. Therefore, the probabilities of genotypes RR, Rr, and rr repectively are 0, 1, and 0, respectively.
This can be veiwed as the following Table: M2W2PT2

Parents' genotypes			Genotypes of offspring
RR-RR	RR-Rr	RR-rr	
1	1/2	0	RR
0	1/2	1	Rr
0	0	0	rr

Table: M2W2PT4

The matrix representing this observation is given by $P = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Let the probabilities of RR, Rr, and rr in the initial (i.e., at $t = 0$) sample space be $X_0^1, X_0^2,$ and X_0^3, respectively. This is represented by the initial distribution vector (3×1 matrix) is denoted by $X_0 = \begin{bmatrix} X_0^1 \\ X_0^2 \\ X_0^3 \end{bmatrix}$.

For any positive integer n, the distribution vector after n generations (i.e., at $t = n$) is denoted by X_n and given by the equation $PX_{n-1} = X_n$.
Using the above information answer the following questions.

8. Find out the correct set of options from the following.
   - ◯ Option 1: The row reduced echelon form of P and P^2 are different in this case.
   - ◯ **Option 2:** The row reduced echelon form of P and P^2 are same in this case.
   - ◯ **Option 3:** The row reduced echelon form of P is $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

◯ Option 4: The row reduced echelon form of P is $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

Solution:

Given $P = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

Applying elementary row transformations to convert P into row reduced echelon form, we get $\xrightarrow{R_1 - R_2} P \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{2R_2} P \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

This is the reduced row echelon form of P.

Now $P^2 = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & \frac{3}{4} & \frac{1}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$.

So $P^2 = \begin{bmatrix} 1 & \frac{3}{4} & \frac{1}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{4R_2} \begin{bmatrix} 1 & \frac{3}{4} & \frac{1}{2} \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - \frac{3}{4}R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

This is the reduced row echelon form of P^2.

Hence the row reduced echelon form of P and P^2 are the same.

Hence options 2 and 3 are correct.

9. Suppose after 2 years the distribution vector i.e., X_2 is calculated to be $\begin{bmatrix} 3/4 \\ 1/4 \\ 0 \end{bmatrix}$ and the initial distribution of RR is $\frac{1}{3}$. Find out the initial distibution of Rr and rr.

◯ **Option 1:** The initial distibution of Rr is : $\frac{1}{3}$.

◯ Option 2: The initial distibution of rr is 0.

◯ **Option 3:** The initial distibutions of Rr and rr are equal.

◯ Option 4: Cannot be determined from the given information.

Solution:

Given $PX_{n-1} = X_n$, and initial distribution of RR is $\frac{1}{3}$.

Let the initial distribution be $X_0 = \begin{bmatrix} \frac{1}{3} \\ y \\ z \end{bmatrix}$.

For simplicity of notation we have considered $X_0^1 = x, X_0^2 = y$ *and* $X_0^3 = z$.

Now $PX_0 = X_1$, and $PX_1 = X_2$

$\implies PPX_0 = PX_1 = X_2$

$\implies P^2X_0 = X_2$

$$\implies \begin{bmatrix} 1 & \frac{3}{4} & \frac{1}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3/4 \\ 1/4 \\ 0 \end{bmatrix}.$$

Now this is a system of linear equations where $x = \frac{1}{3}$.

Observe that we can not apply the Cramer's rule as the determinant of the coefficient matrix is zero.

Now the augmented matrix of the above system of linear equations is

$$[A|B] = \left[\begin{array}{ccc|c} 1 & \frac{3}{4} & \frac{1}{2} & \frac{3}{4} \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{4R_2} \left[\begin{array}{ccc|c} 1 & \frac{3}{4} & \frac{1}{2} & \frac{3}{4} \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 - \frac{3}{4}R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

So from reduced row echelon form of augmented matrix we have $x - z = 0$.

$\implies \quad z = \frac{1}{3}$ as $x = \frac{1}{3}$,

and $y + 2z = 1 \quad \implies \quad y + \frac{2}{3} = 1 \quad \implies y = \frac{1}{3}$.

But y, z denote the initial distribution of Rr and rr respectively.

Hence the initial distribution of Rr and rr are $\frac{1}{3}$ and $\frac{1}{3}$ respectively.

Hence options 1 and 3 are correct.

10. Suppose after 3 generations the distribution vector i.e., X_3 is calculated to be $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, and recall that $0 \leq X_0^1, X_0^2, X_0^3 \leq 1$. Find out the correct set of options.

- ◯ **Option 1:** $X_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
- ◯ Option 2: $X_0 = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$
- ◯ Option 3: $X_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
- ◯ Option 4: X_0 cannot be determined from the given information.

◯ **Option 5:** $X_0 = X_n$ for all positive integer n.

◯ Option 6: There can be some positive integer n for which $X_0 \neq X_n$.

Solution:

Given $PX_{n-1} = X_n$ and $X_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. $\implies PX_0 = X_1,\ PX_1 = X_2\ and\ PX_2 = X_3$

$\implies P^2X_0 = PX_1 = X_2$

$\implies P^3X_0 = P^2X_1 = PX_2 = X_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Now $P^3 = P^2.P = \begin{bmatrix} 1 & \frac{3}{4} & \frac{1}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & \frac{7}{8} & \frac{3}{4} \\ 0 & \frac{1}{8} & \frac{1}{4} \\ 0 & 0 & 0 \end{bmatrix}$.

For simplicity of notation we have consider $X_0^1 = x, X_0^2 = y$ and $X_0^3 = z$.

So the initial distribution $X_0 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

$P^3X_0 = X_3 \implies \begin{bmatrix} 1 & \frac{7}{8} & \frac{3}{4} \\ 0 & \frac{1}{8} & \frac{1}{4} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

The augmented matrix for the above system of equations is

$$[A|B] = \left[\begin{array}{ccc|c} 1 & \frac{7}{8} & \frac{3}{4} & 1 \\ 0 & \frac{1}{8} & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{8R_2} \left[\begin{array}{ccc|c} 1 & \frac{7}{8} & \frac{3}{4} & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 - 7/8R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This is the row reduced echelon form of a augmented matrix, so

$$x - z = 1 \tag{1}$$

$$y + 2z = 0 \tag{2}$$

$\implies x = 1 + z, y = -2z.$

It is given that $0 \leq x, y, z \leq 1$, so from equation (1) we get $x \geq 1\ as\ z \geq 0$ but also we have $x \leq 1 \implies x = 1$.

Hence from equation (1) $z = 0$ so from equation (2) $y = 0$.

So $X_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Now $PX_{n-1} = X_n \implies PX_0 = X_1 \implies X_1 = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Similarly $X_2 = PX_1 = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

and so on, we get $X_n = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Hence options 1 and 5 are correct.